

2.2 Equations

Subject content	Guidance	Curriculum reference
A Find pairs of numbers that satisfy a linear equation with two unknowns	$x + y = 7$ Write down 2 possible values for x and y	A8.2A
B Solve linear equations including unknowns on both sides	Solve $4x - 3 = 2x + 7$ $5y + 4 = 7y + 5$	A8.2B A9.2A
C Solve linear equations, including unknowns on both sides and/or include brackets or fractions	Solve $\frac{5(x-3)}{4} = \frac{2x+6}{2}$	A8.2B A8.2C A9.2B
D Solve problems by writing and solving equations	A man said 'My brother is 8 years' older than me, my sister is 5 years' younger than me, and all three of our ages add up to 129'. How old is the man's brother?	A8.2D
E Solve equations involving an x^2 term and a number	Solve $x^2 - 25 = 96$	A9.2C
F Solve quadratic equations of the form $x^2 + bx + c = 0$, by factorising	Solve $x^2 - 7x - 30 = 0$	A9.2D

2.2 Equations *continued*

Subject content	Guidance	Curriculum reference
G Write equations to represent direct proportion	When $x=16$, $y=4$ y is directly proportional to x Write a proportionality equation involving x and y	A9.2E
H Solve a pair of simultaneous linear equations	Solve the simultaneous equations: $2x + 3y = -4$ $4x - y = 6$	A9.2F
I Solve problems involving simultaneous linear equations or direct proportion	4 burgers and 3 cakes costs \$14.80 5 burgers and 2 cakes costs \$15.70 How much would 6 burgers and 4 cakes cost?	A9.2G